Diary Data Analysis

Anne Gaule

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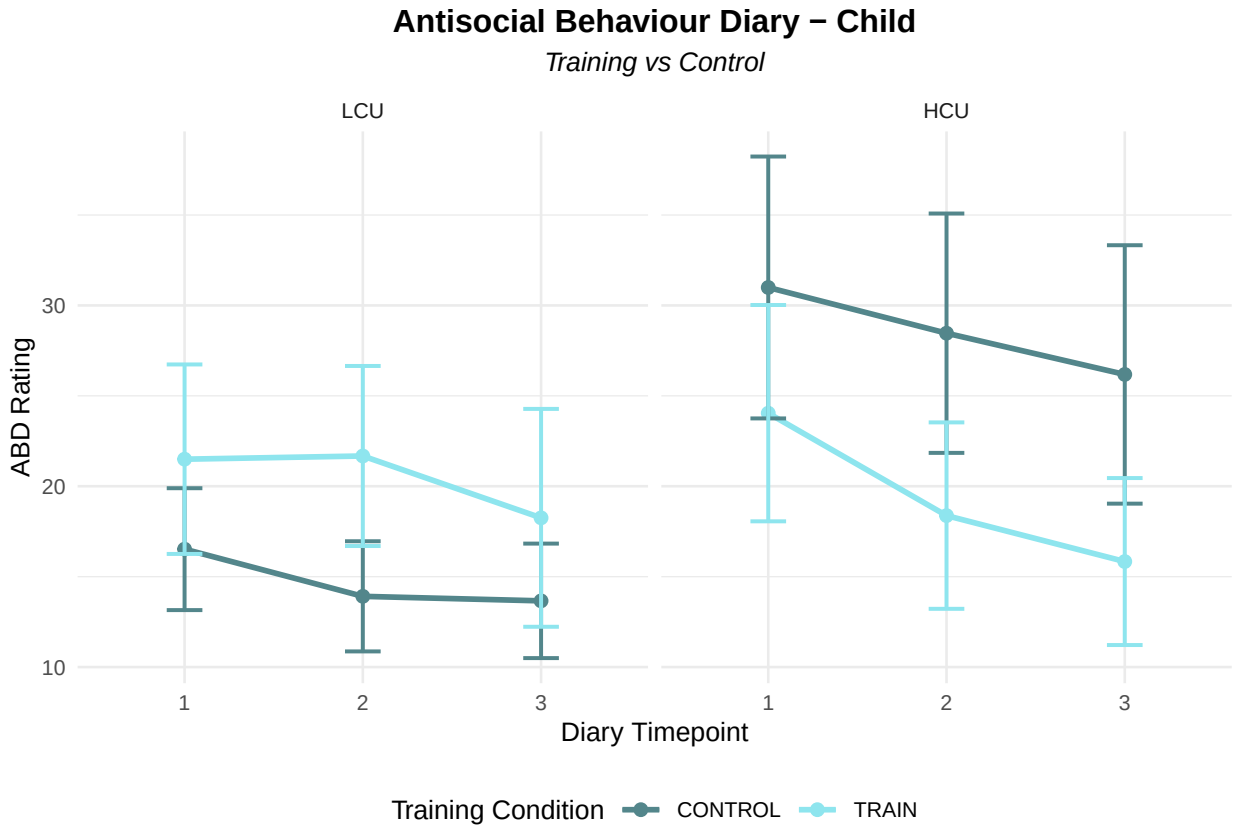
1) Child rated ASB at 2 months - by CU

1.1) Table of outcomes

Training_Condition	diaryTP	CP_Group	Group_N_ABD	Mean_child_rated_ABD	se_child_rated_ABD	NA_child_rated_ABD
CONTROL	1	LCU	26	16.52077	3.370843	0
CONTROL	1	HCU	18	30.99333	7.242612	0
CONTROL	2	LCU	26	13.91154	3.045668	0
CONTROL	2	HCU	18	28.46444	6.619444	0
CONTROL	3	LCU	22	13.66136	3.166412	0
CONTROL	3	HCU	17	26.18353	7.146762	0
TRAIN	1	LCU	22	21.49636	5.240413	0
TRAIN	1	HCU	16	24.04125	5.982017	0
TRAIN	2	LCU	22	21.67318	4.978893	0
TRAIN	2	HCU	16	18.37625	5.155261	0
TRAIN	3	LCU	18	18.25222	6.024783	0
TRAIN	3	HCU	15	15.83533	4.618226	0

1.2) Plot of outcomes

First plotting data:



1.3) Linear Regression

```
lm_model_c_time_asb_d3_only <- lm(A_sum_child ~ ABD_sum_D1 +
                          Training_Condition*CP_Group,
                        data =  child_abd_d3_only)



# model without interaction
lm_model_c_time_asb_d3_only_nointeraction <- lm(A_sum_child ~ ABD_sum_D1 +
                          Training_Condition + CP_Group,
                        data =  child_abd_d3_only)
```

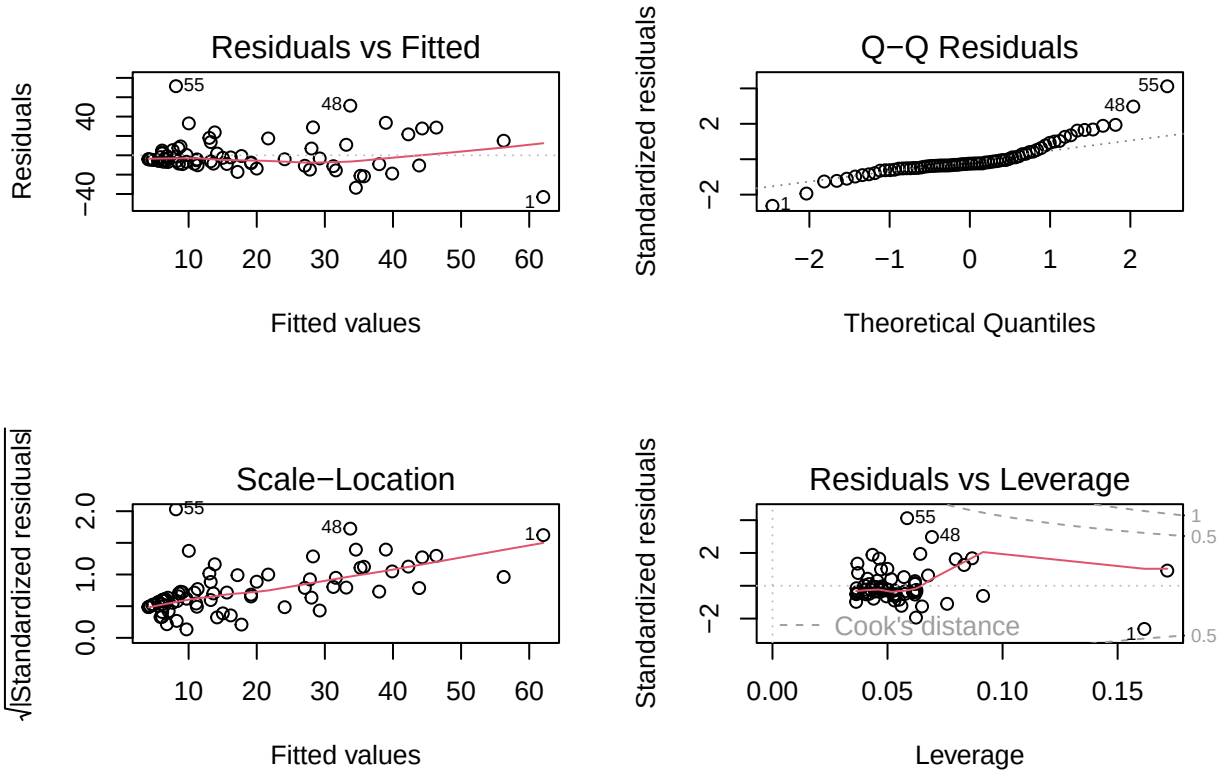
1.3.1) Does interaction term improve model? No.

```
## Analysis of Variance Table
##
## Model 1: A_sum_child ~ ABD_sum_D1 + Training_Condition + CP_Group
## Model 2: A_sum_child ~ ABD_sum_D1 + Training_Condition * CP_Group
##   Res.Df   RSS Df Sum of Sq      F Pr(>F)
## 1     68 21709
## 2     67 21177  1    531.69 1.6822 0.1991
```

1.3.2) Model results

```
##
## Call:
## lm(formula = A_sum_child ~ ABD_sum_D1 + Training_Condition +
##     CP_Group, data = child_abd_d3_only)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -43.063  -8.579  -4.360   5.000  71.409
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)                4.3555     3.8999   1.117    0.268
## ABD_sum_D1                 0.5925     0.0915   6.476 1.23e-08 ***
## Training_ConditionTRAIN   -0.2503     4.2362  -0.059    0.953
## CP_GroupHCU                2.0081     4.2754   0.470    0.640
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 17.87 on 68 degrees of freedom
##   (2 observations deleted due to missingness)
## Multiple R-squared:  0.3925, Adjusted R-squared:  0.3656
## F-statistic: 14.64 on 3 and 68 DF,  p-value: 1.866e-07
```

1.3.3) Assumption tests



1.4) Bootstrapped regression

1.4.1) Specification and results

```
boot_coef_c_asb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(A_sum_child ~ ABD_sum_D1 + Training_Condition + CP_Group,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
## Boot function: Generates R bootstrap replicates of a statistic applied to data.
## data = my data
## statistic = A function which, when applied to data returns a vector containing the statistic(s) of i
set.seed(123)  # For c
boot_results_c_asb <- boot(data = child_abd_d3_only,
                     statistic = boot_coef_c_asb,
                     R = 5000)

# View results
boot_results_c_asb
```

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = child_abd_d3_only, statistic = boot_coef_c_asb, R = 5000)
##
##
## Bootstrap Statistics :
##       original       bias    std. error
## t1*  4.3555245 -0.028695643   3.3142693
## t2*  0.5925422  0.002106226   0.1266314
## t3* -0.2502926 -0.136901625   4.2205987
## t4*  2.0081172  0.150158702   4.4151847
```

1.4.2) Confidence Intervals

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   (-1.284, 11.948 )
## Calculations and Intervals on Original Scale

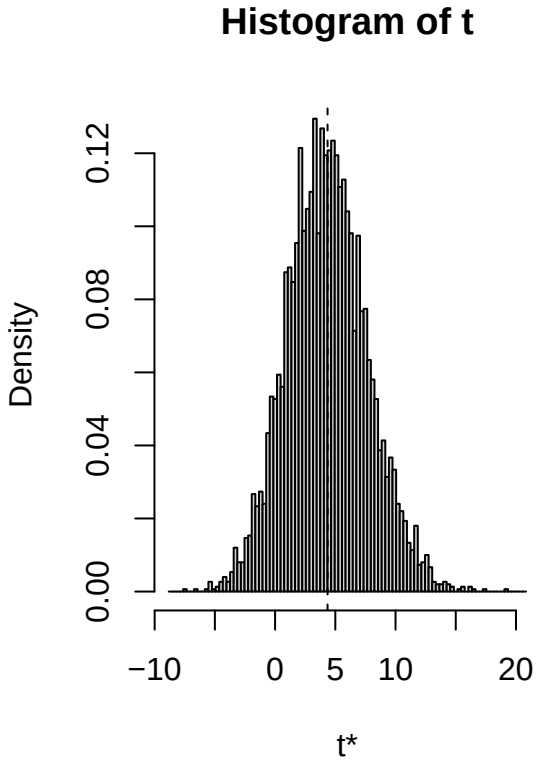
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.3207,  0.8168 )
## Calculations and Intervals on Original Scale

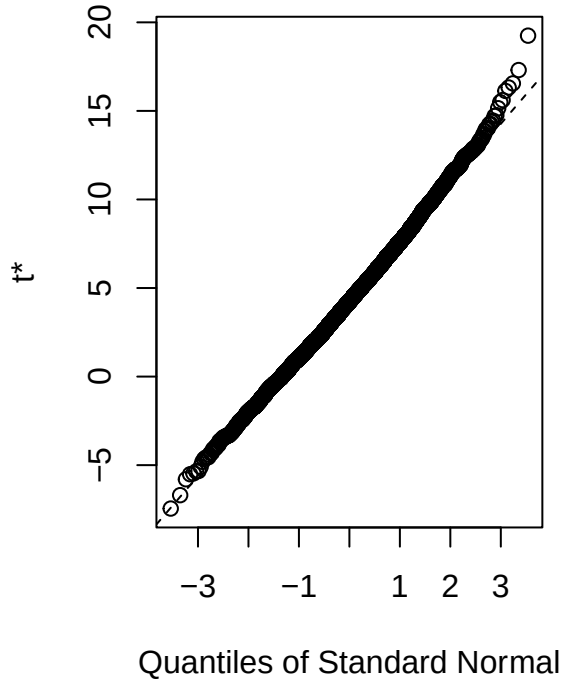
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-9.0744,  7.7372 )
## Calculations and Intervals on Original Scale

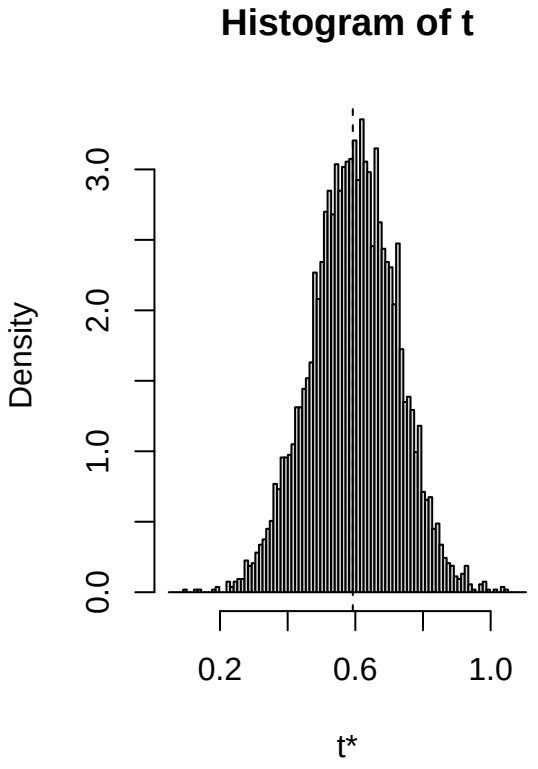
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
```

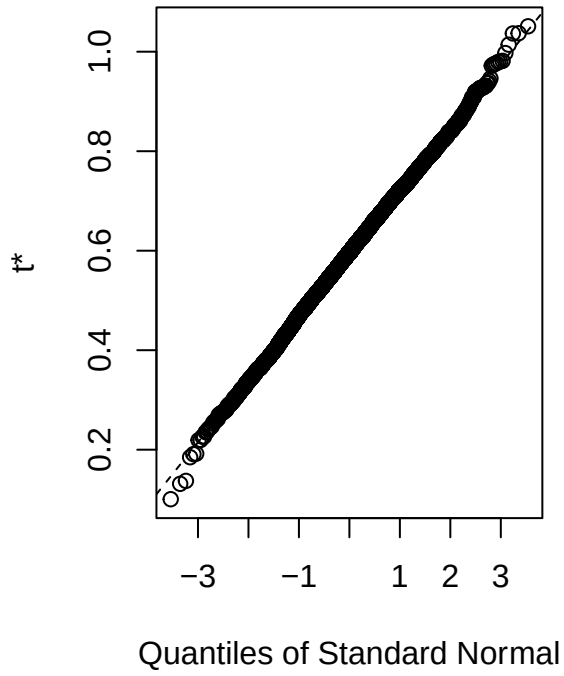
```
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-6.021, 11.735 )
## Calculations and Intervals on Original Scale
```

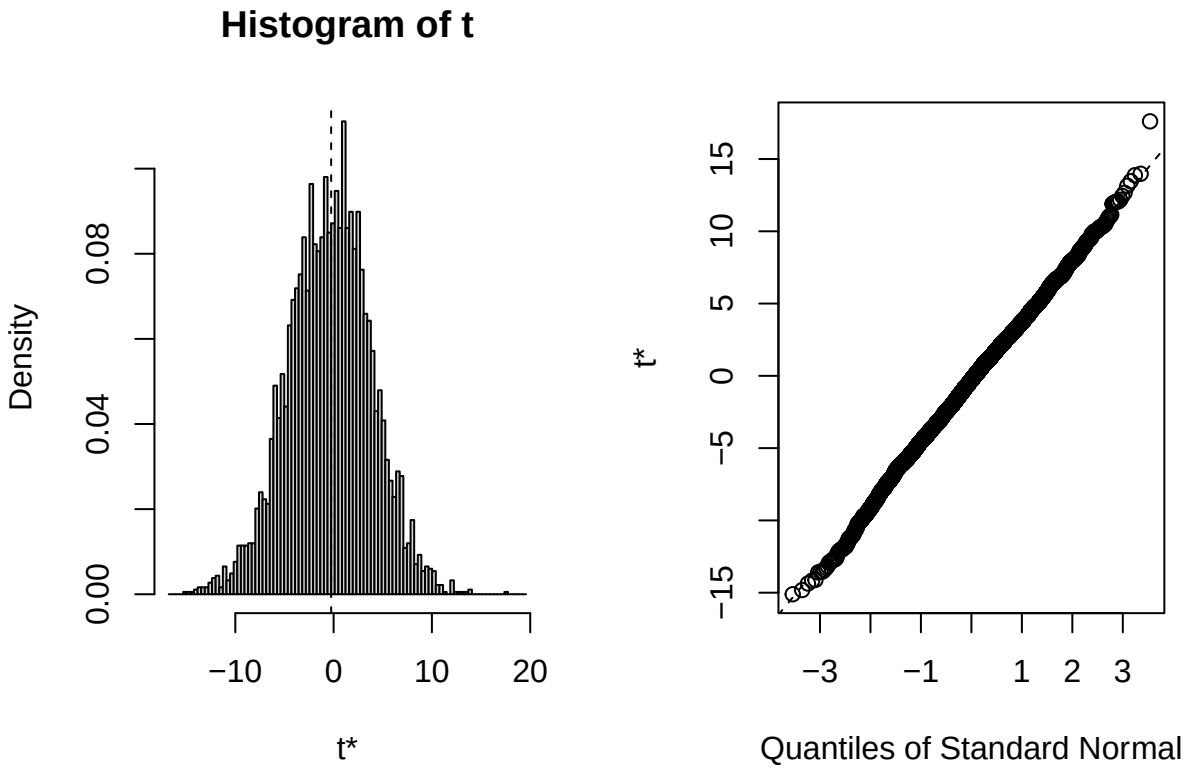
1.4.3) Model diagnostics

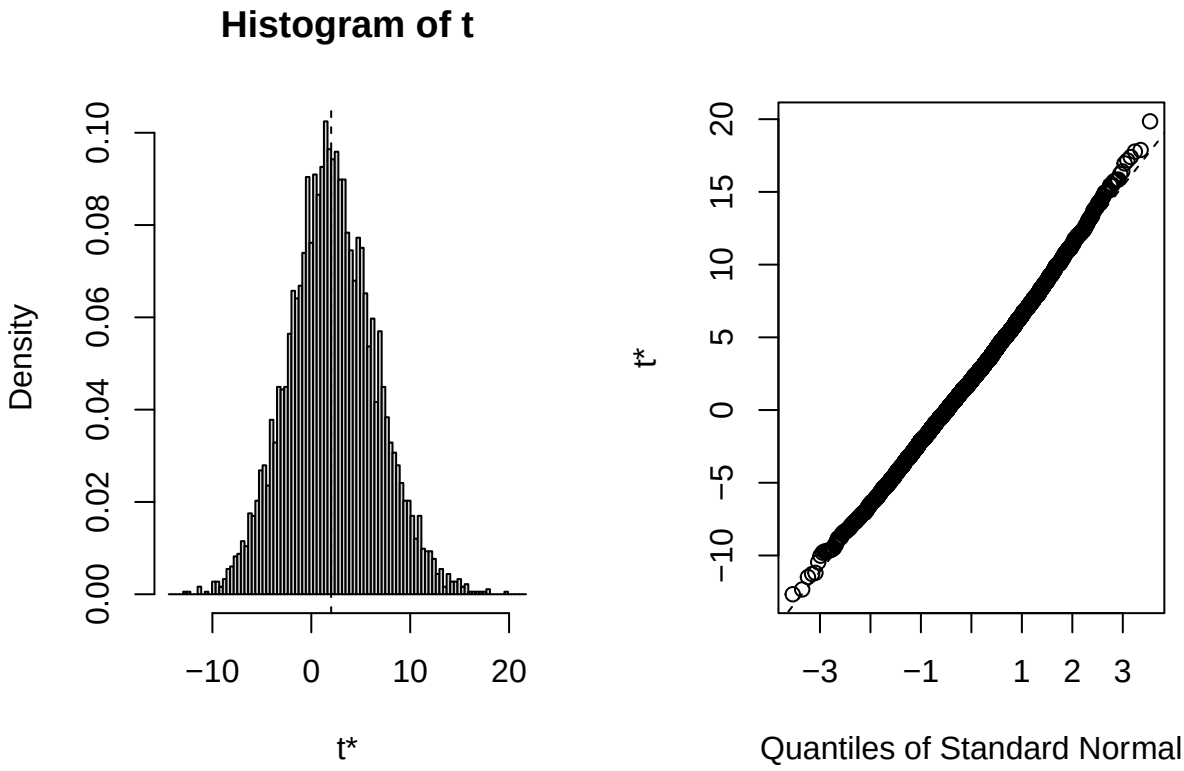









Histogram of t
Density
0.00
0.04
0.08
−10
0
10
20
t*
t*
−15
−5
0
5
10
15
−3
−1
1
2
3
Quantiles of Standard Normal

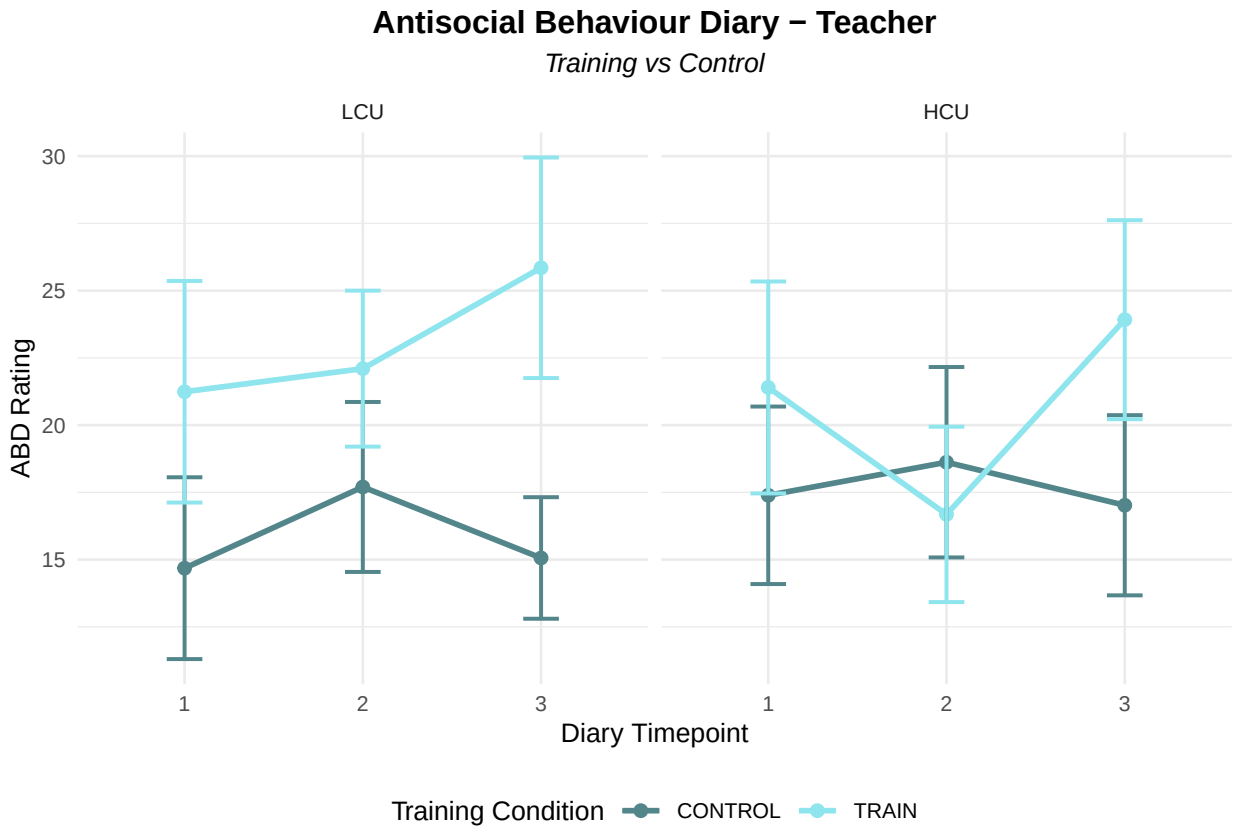

Histogram of t
Density
0.00
0.02
0.04
0.06
0.08
0.10
−10
0
10
20
t*
t*
−10
−5
0
5
10
15
20
−3
−1
1
2
3
Quantiles of Standard Normal

2) Teacher rated ASB at two months by CU

2.1) Table of outcomes

Training_Condition	diaryTP	CP_Group	Group_N	Mean_ABD	se_ABD	Mean_PBD	se_PBD
CONTROL	1	LCU	32	14.68	3.38	37.49	4.023299
CONTROL	1	HCU	21	17.39	3.30	18.08	3.305231
CONTROL	2	LCU	32	17.70	3.16	35.24	3.616988
CONTROL	2	HCU	21	18.62	3.54	24.13	4.220790
CONTROL	3	LCU	30	15.06	2.26	34.88	4.146050
CONTROL	3	HCU	21	17.02	3.35	28.66	4.017480
TRAIN	1	LCU	25	21.24	4.12	39.55	4.566927
TRAIN	1	HCU	25	21.40	3.94	27.64	3.650338
TRAIN	2	LCU	25	22.10	2.90	38.41	4.598401
TRAIN	2	HCU	25	16.68	3.26	30.15	3.809410
TRAIN	3	LCU	21	25.85	4.10	31.14	5.258720
TRAIN	3	HCU	24	23.92	3.70	23.94	3.811794

2.2) Plot of outcomes



2.3) Linear Regression

```
## lmer gives singular fit so trying simple lm model
model_t_time_asb_nod2_lm<- lm(ABD_mean_rating ~ Mean_ABD_D1 +
                          Training_Condition*CP_Group,
                      data =  teach_abd_d3_only)

# model without interaction
model_t_asb_CU_lm_nointeraction  <- lm(ABD_mean_rating ~ Mean_ABD_D1 +
                          Training_Condition + CP_Group,
                      data =  teach_abd_d3_only)
```

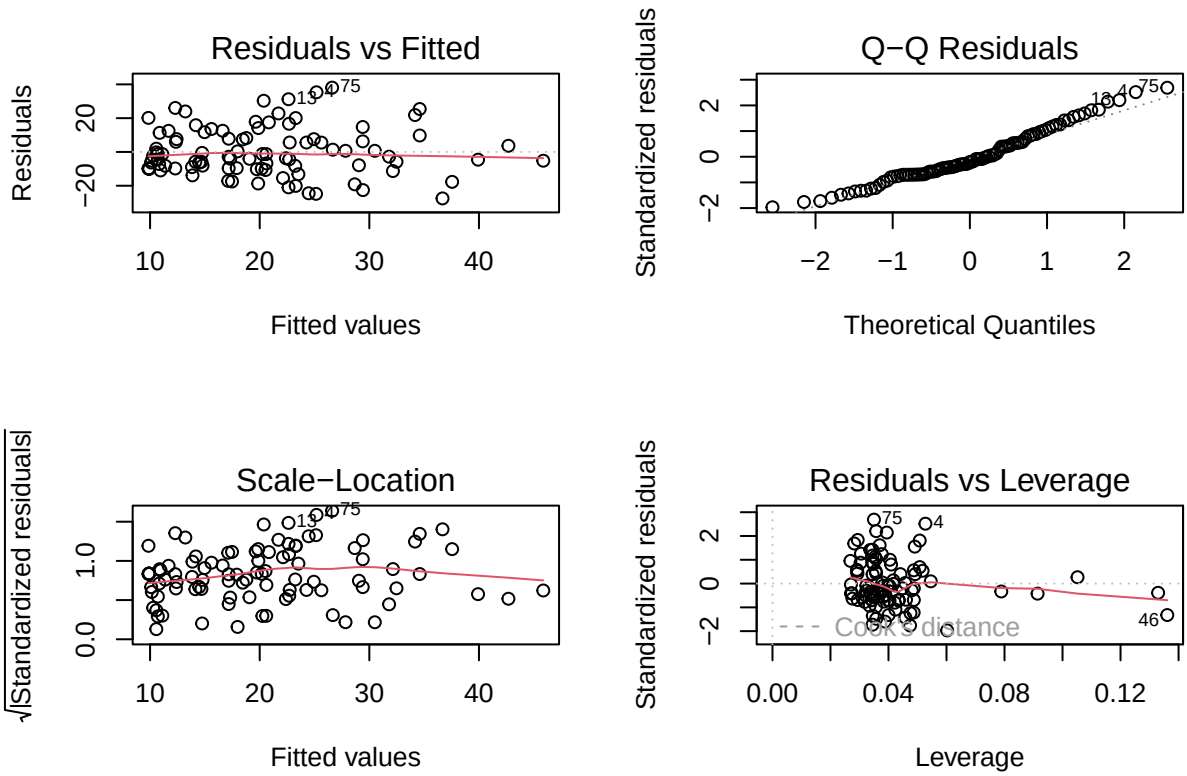
2.3.1) Does interaction term improve fit? no.

```
## Analysis of Variance Table
##
## Model 1: ABD_mean_rating ~ Mean_ABD_D1 + Training_Condition * CP_Group
## Model 2: ABD_mean_rating ~ Mean_ABD_D1 + Training_Condition + CP_Group
##   Res.Df   RSS Df Sum of Sq      F Pr(>F)
## 1     90 18862
## 2     91 18903 -1   -41.531 0.1982 0.6573
```

2.3.2) Model results

```
##
## Call:
## lm(formula = ABD_mean_rating ~ Mean_ABD_D1 + Training_Condition +
##     CP_Group, data = teach_abd_d3_only)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -27.511  -9.826  -2.837   8.024  38.061
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               9.85344    2.67486   3.684  0.00039 ***
## Mean_ABD_D1               0.36689    0.07970   4.603 1.34e-05 ***
## Training_ConditionTRAIN  7.27460    3.02001   2.409  0.01802 *
## CP_GroupHCU               0.04278    2.98623   0.014  0.98860
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.41 on 91 degrees of freedom
## Multiple R-squared:  0.2569, Adjusted R-squared:  0.2324
## F-statistic: 10.48 on 3 and 91 DF,  p-value: 5.451e-06
```

2.3.3) Assumption checks



2.4) Bootstrapped linear regression

2.4.1) Specification and results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = teach_abd_d3_only, statistic = boot_coef_t_asb_CU,
##     R = 5000)
##
##
## Bootstrap Statistics :
##      original       bias    std. error
## t1* 9.8534351 -0.080240668  2.35166390
## t2* 0.3668917  0.006087193  0.07807368
## t3* 7.2746033 -0.076803660  3.12277373
## t4* 0.0427813 -0.007811426  2.97897518
```

2.4.2) Confidence intervals

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_CU, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   ( 5.582, 14.836 )
## Calculations and Intervals on Original Scale

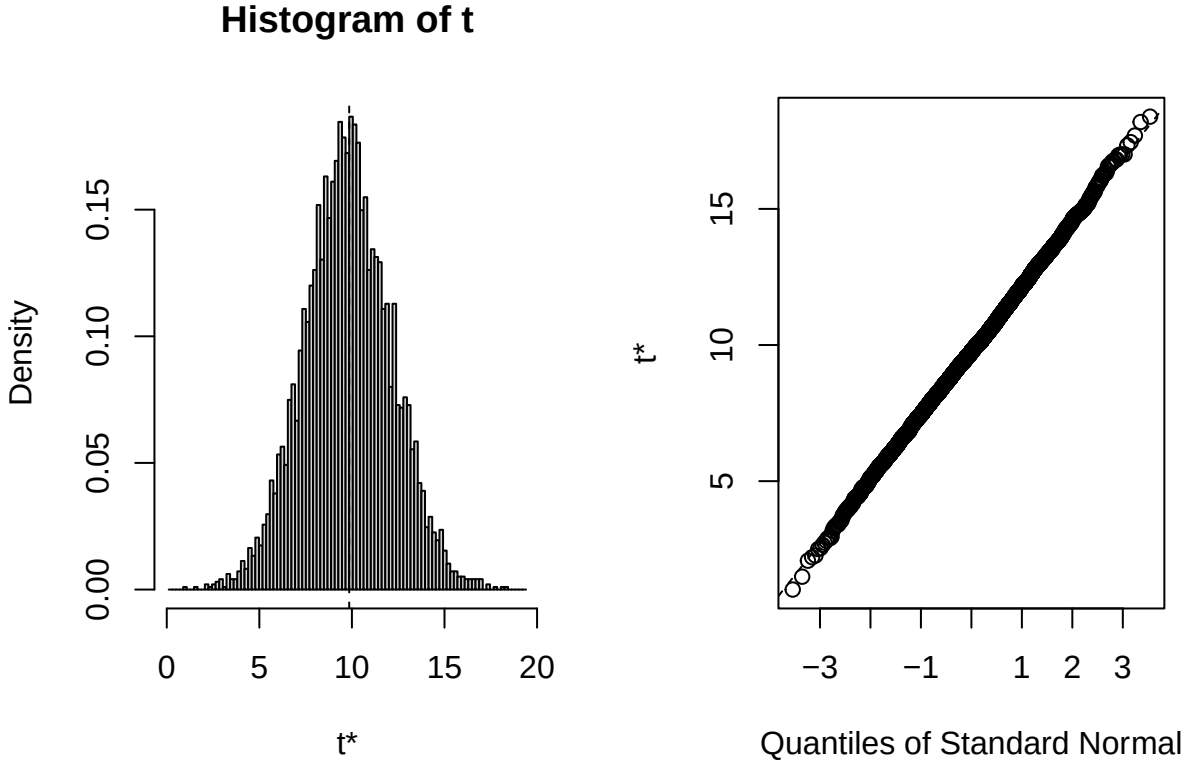
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_CU, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.2193,  0.5242 )
## Calculations and Intervals on Original Scale

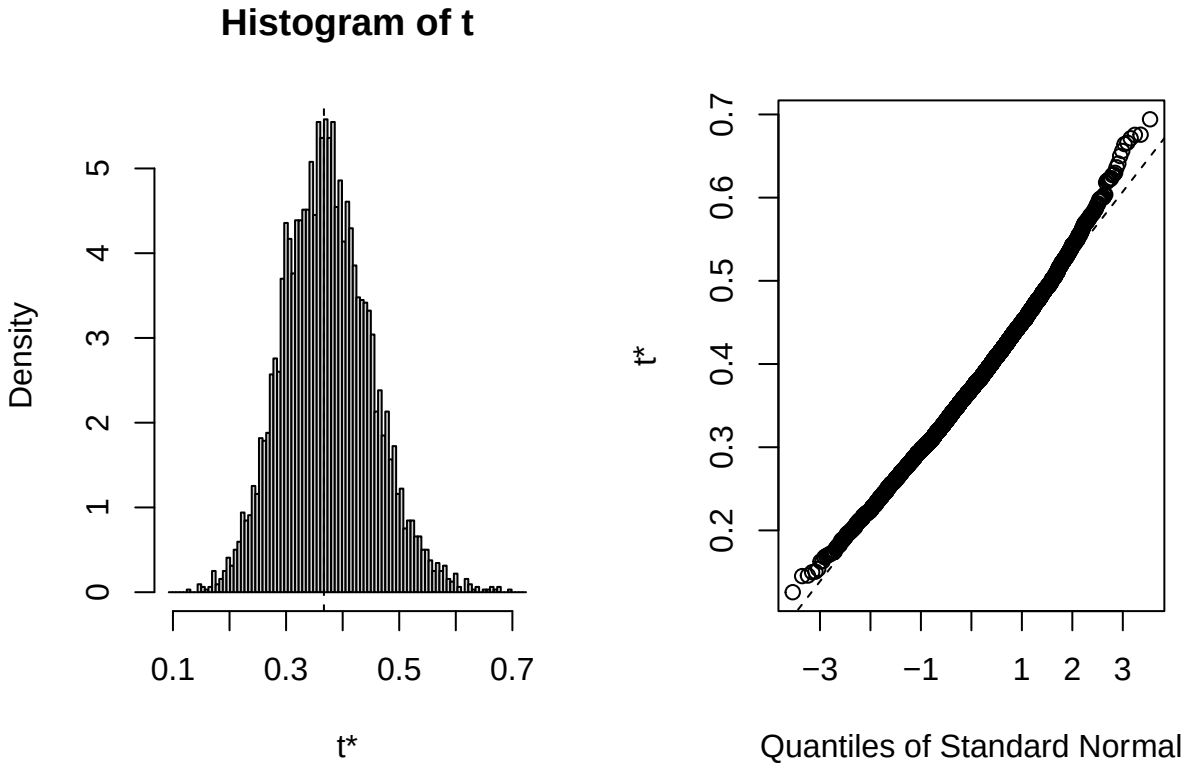
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_CU, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   ( 1.567, 13.824 )
## Calculations and Intervals on Original Scale
```

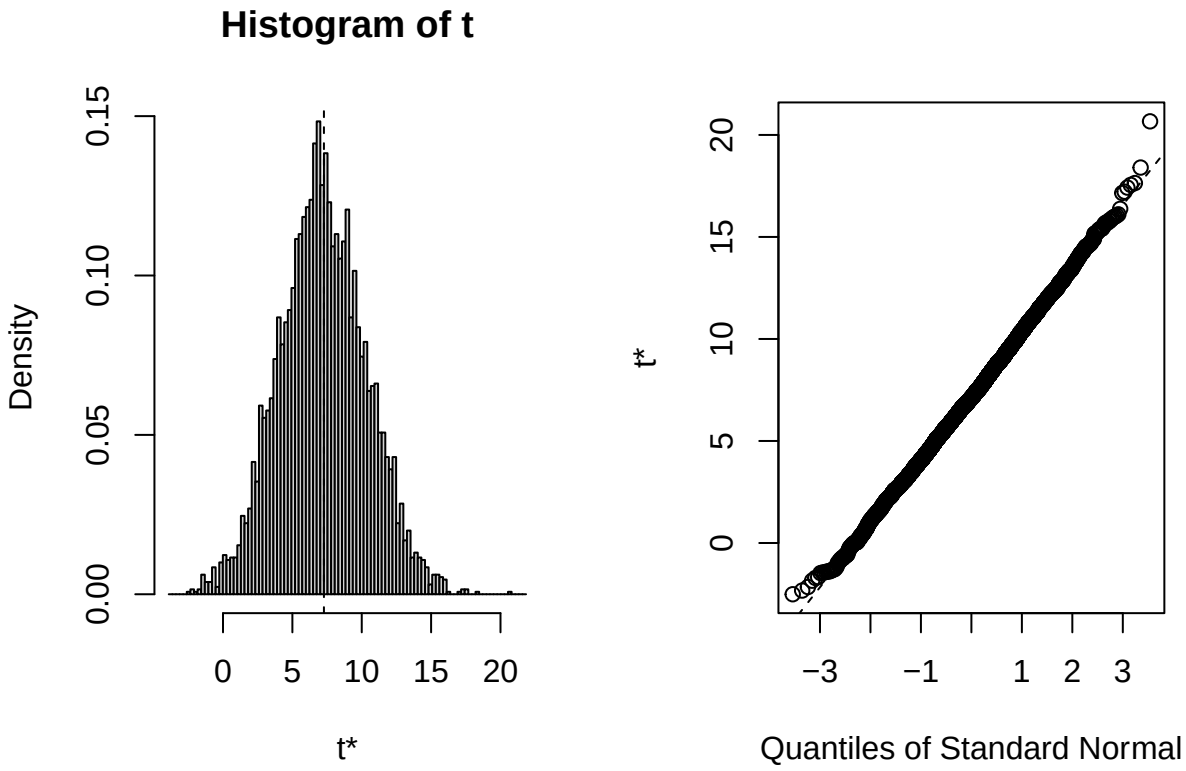
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_CU, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-5.8371,  5.9010 )
## Calculations and Intervals on Original Scale
```

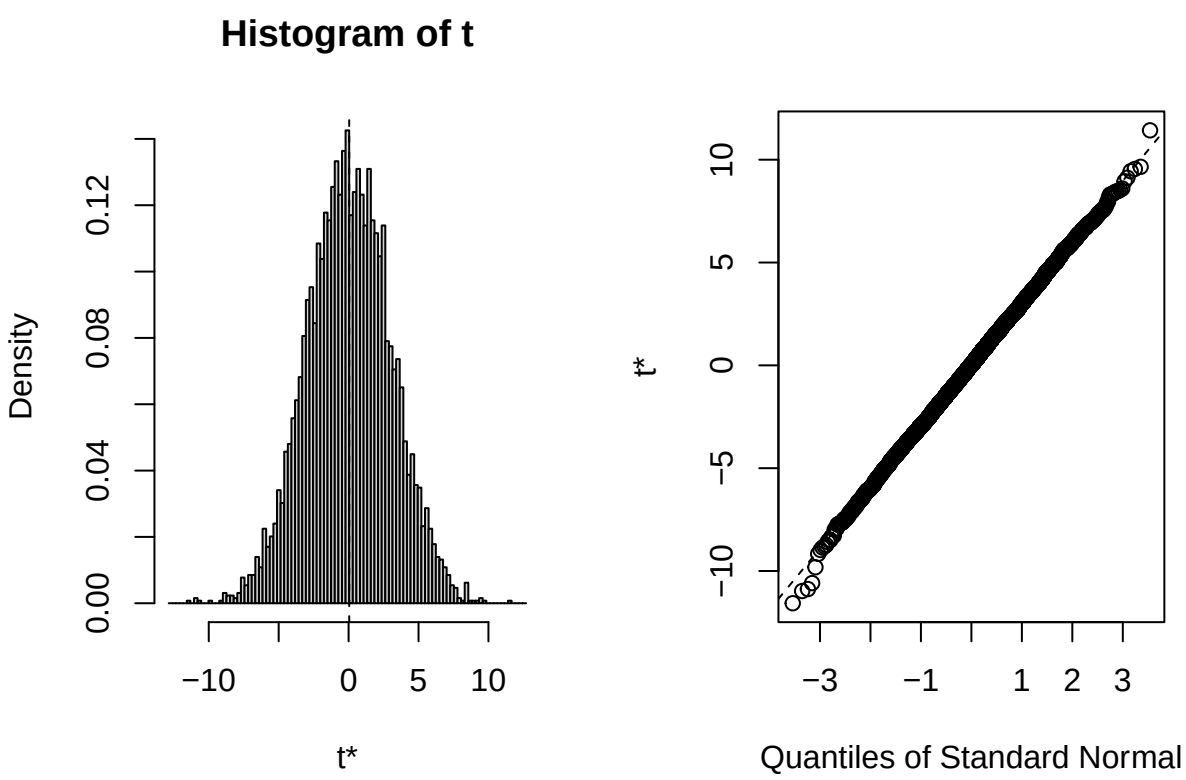
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_CU, conf = 0.99, type = "bca",
##     index = 3)
##
## Intervals :
## Level       BCa
## 99%   (-0.413, 15.750 )
## Calculations and Intervals on Original Scale
```

2.4.3) Model diagnostics






Histogram of t
Density
0.00
0.05
0.10
0.15
0
5
10
15
20
t*
t*
0
5
10
15
20
−3
−1
1
2
3
Quantiles of Standard Normal




```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_CU, conf = 0.99, type = "bca",
##     index = 1)
##
## Intervals :
## Level       BCa
## 99%   ( 4.144, 16.740 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_CU, conf = 0.99, type = "bca",
##     index = 2)
##
## Intervals :
## Level       BCa
## 99%   ( 0.1733,  0.5816 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
```

```
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_CU, conf = 0.99, type = "bca",
##     index = 3)
##
## Intervals :
## Level       BCa
## 99%   (-0.413, 15.750 )
## Calculations and Intervals on Original Scale

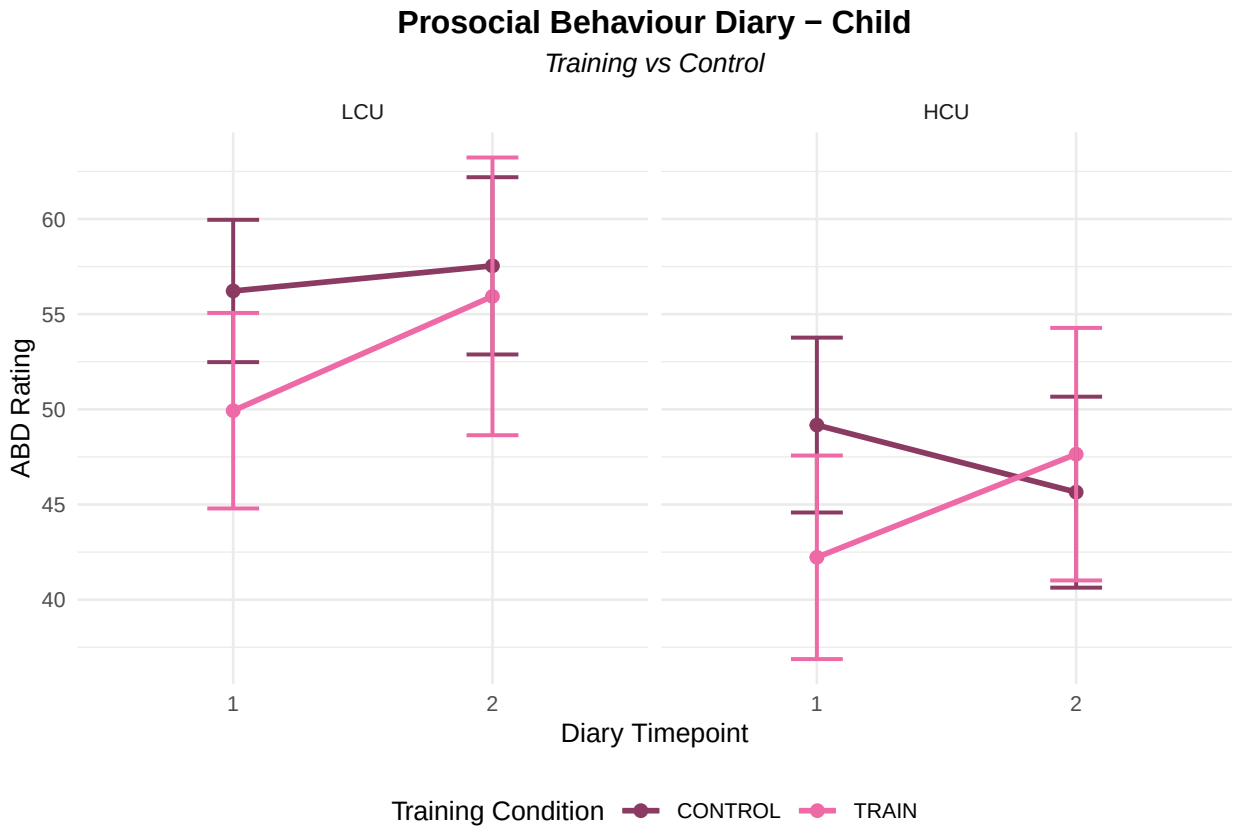
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_CU, conf = 0.99, type = "bca",
##     index = 4)
##
## Intervals :
## Level       BCa
## 99%   (-7.6551,  7.5234 )
## Calculations and Intervals on Original Scale
```

3) Child Prosocial at two weeks

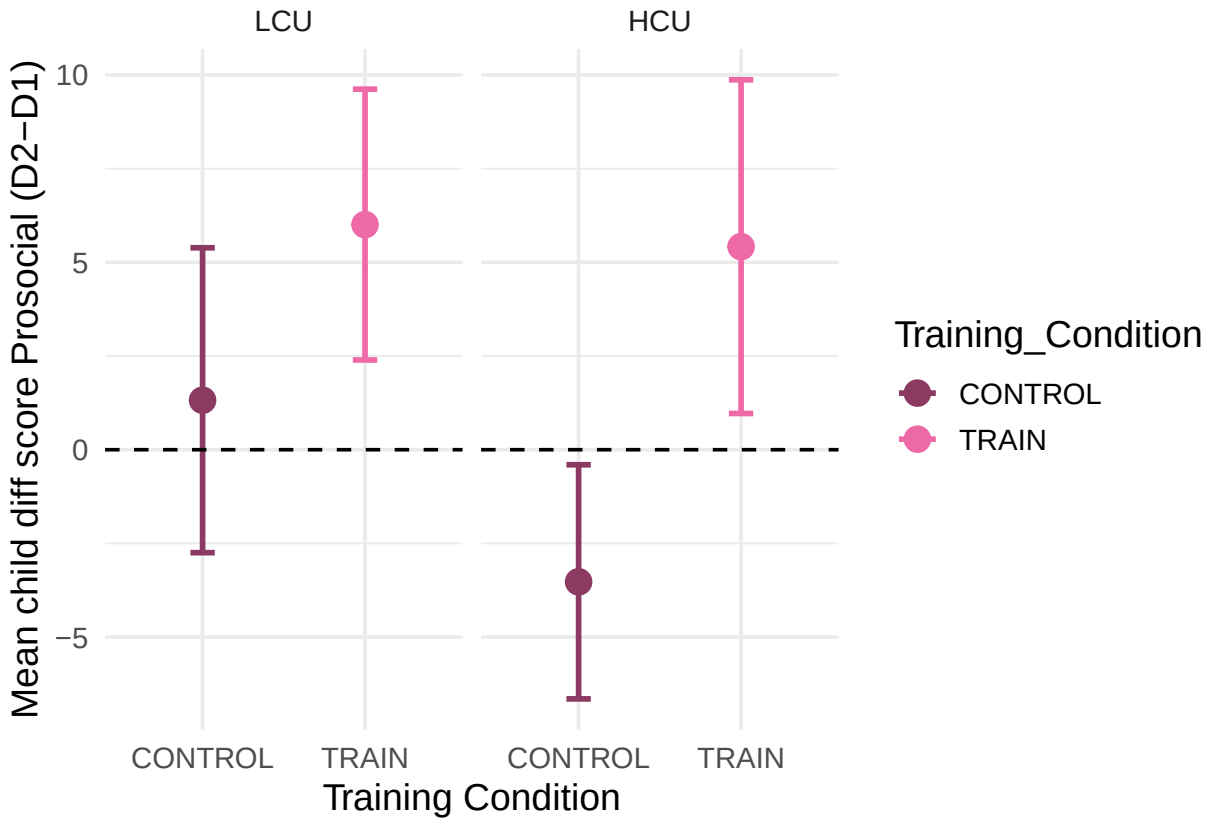
3.1) Table of outcomes

Training_Condition	diaryTP	CP_Group	Group_N_ABD	Mean_child_rated_ABD	se_child_rated_ABD	NA_child_rated_ABD
CONTROL	1	LCU	26	16.52077	3.370843	0
CONTROL	1	HCU	18	30.99333	7.242612	0
CONTROL	2	LCU	26	13.91154	3.045668	0
CONTROL	2	HCU	18	28.46444	6.619444	0
CONTROL	3	LCU	22	13.66136	3.166412	0
CONTROL	3	HCU	17	26.18353	7.146762	0
TRAIN	1	LCU	22	21.49636	5.240413	0
TRAIN	1	HCU	16	24.04125	5.982017	0
TRAIN	2	LCU	22	21.67318	4.978893	0
TRAIN	2	HCU	16	18.37625	5.155261	0
TRAIN	3	LCU	18	18.25222	6.024783	0
TRAIN	3	HCU	15	15.83533	4.618226	0

3.2) Plot of outcomes



Difference plot:


LCU
HCU
Mean child diff score Prosocial (D2−D1)
10
5
0
−5
CONTROL
TRAIN
CONTROL
TRAIN
Training Condition
Training_Condition
CONTROL
TRAIN

3.3) Linear regression

```
# singular fit so simplfying

model_c_psb_CU_lm <- lm(PSB_mean_D2 ~ PSB_mean_D1 + Training_Condition + CP_Group +
                          Training_Condition*CP_Group,
                        data =  teach_psb_d1d2_wide)


# model without interaction
model_c_psb_CU_lm_nointeraction <- lm(PSB_mean_D2 ~ PSB_mean_D1 + Training_Condition + CP_Group,
                        data =  teach_psb_d1d2_wide)
```

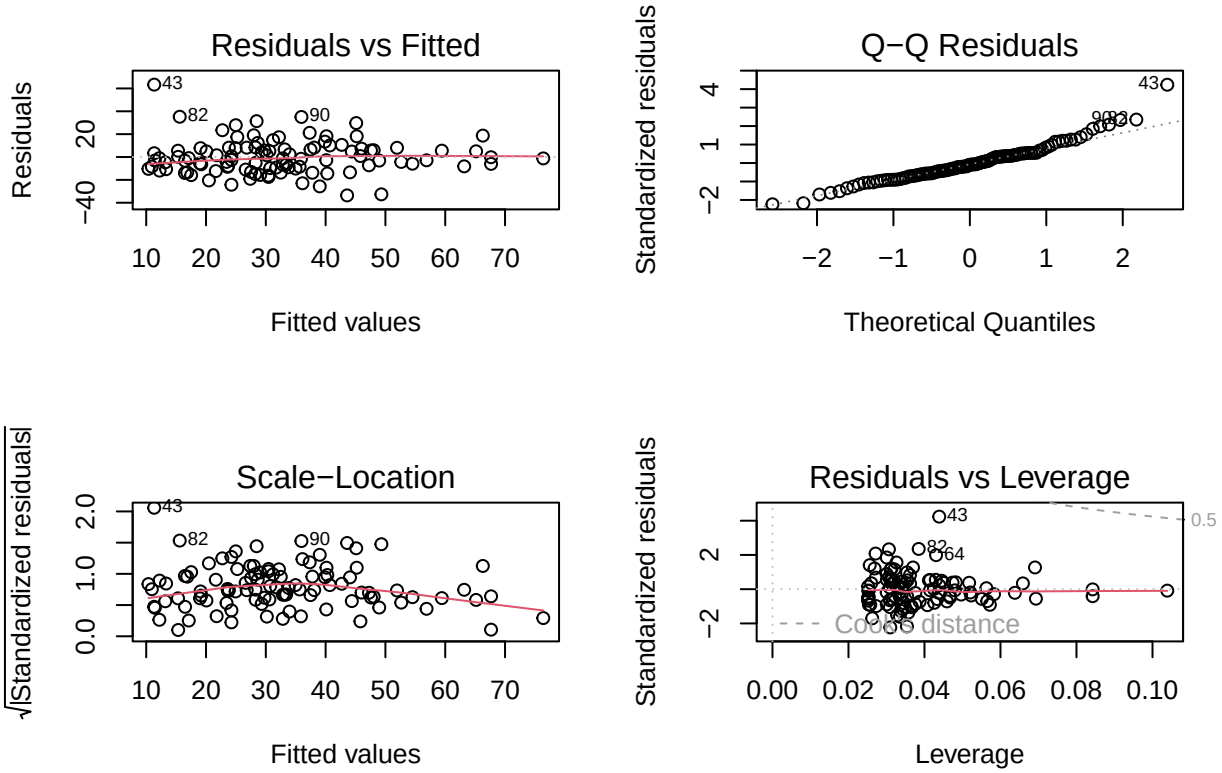
3.3.1) Does interaction term improve fit?

```
## Analysis of Variance Table
##
## Model 1: PSB_mean_D2 ~ PSB_mean_D1 + Training_Condition + CP_Group + Training_Condition *
##     CP_Group
## Model 2: PSB_mean_D2 ~ PSB_mean_D1 + Training_Condition + CP_Group
##   Res.Df   RSS Df Sum of Sq     F Pr(>F)
## 1     97 22866
## 2     98 22892 -1   -26.406 0.112 0.7386
```

3.3.2) Model summary

```
##
## Call:
## lm(formula = PSB_mean_D2 ~ PSB_mean_D1 + Training_Condition +
##     CP_Group, data = teach_psb_d1d2_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -33.596  -9.616  -1.385   7.618  63.332
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)              10.37655    3.63357   2.856  0.00524 **
## PSB_mean_D1               0.67397    0.07555   8.921 2.65e-14 ***
## Training_ConditionTRAIN  0.86896    3.07457   0.283  0.77806
## CP_GroupHCU               0.95821    3.26973   0.293  0.77010
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.28 on 98 degrees of freedom
## Multiple R-squared:  0.4801, Adjusted R-squared:  0.4642
## F-statistic: 30.16 on 3 and 98 DF,  p-value: 6.712e-14
```

3.3.3) Assumption checks



3.4) Bootstrapped linear regression

3.4.1) Results and specification

```
# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
## 1) first resample from my dataframe (with replacement)
## 2) next fit model, but with my resampled dataframe.
## 3) return 95%cis for all model coefficients
boot_coef_t_asb_CU <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(P_sum_child ~ PBD_sum_D1 +
                          Training_Condition + CP_Group,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
## Boot function: Generates R bootstrap replicates of a statistic applied to data.
## data = my data
## statistic = A function which, when applied to data returns a vector containing the statistic(s) of i
set.seed(123)  # For reproducibility
boot_results_t_asb_CU <- boot(data = child_pbd_d3_only,
                     statistic = boot_coef_t_asb_CU,
                     R = 5000)

# View results
boot_results_t_asb_CU
```

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = child_pbd_d3_only, statistic = boot_coef_t_asb_CU,
##     R = 5000)
##
##
## Bootstrap Statistics :
##        original        bias    std. error
## t1* 10.40587004 -0.0223320766   6.3918404
## t2*  0.71400996  0.0003774123   0.1095131
## t3*  4.69024568  0.0196656982   4.6531757
## t4* -0.02120968  0.0444885951   4.6184939
```

4.3.2) Confidence Intervals

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_CU, type = "bca", index = 1)
```

```
##
## Intervals :
## Level       BCa
## 95%   (-1.43, 23.81 )
## Calculations and Intervals on Original Scale

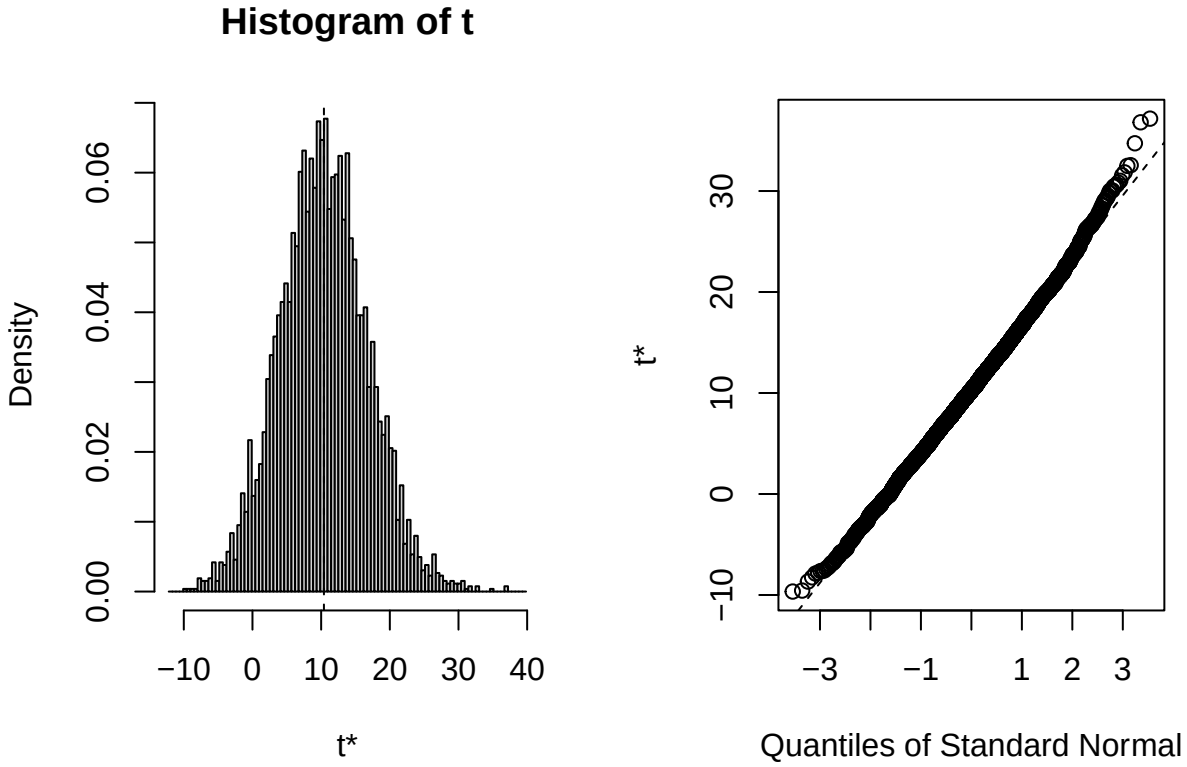
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_CU, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.4885,  0.9121 )
## Calculations and Intervals on Original Scale

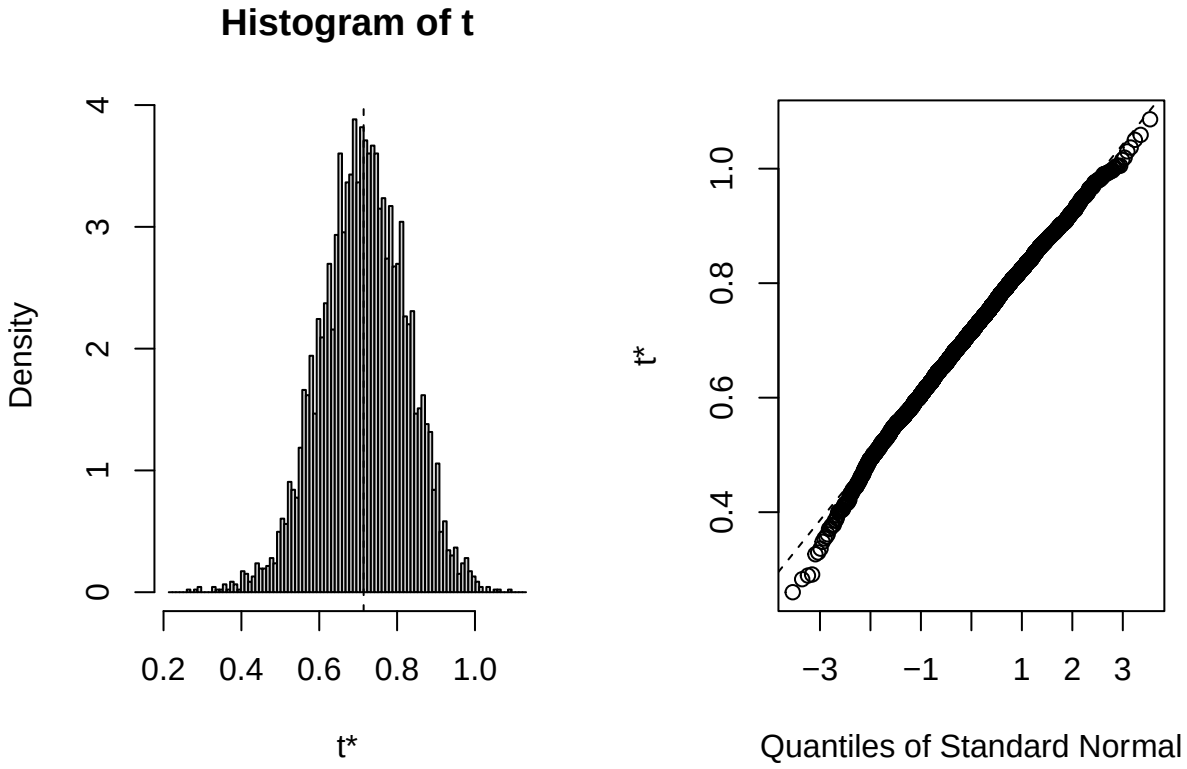
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_CU, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-4.211, 13.998 )
## Calculations and Intervals on Original Scale

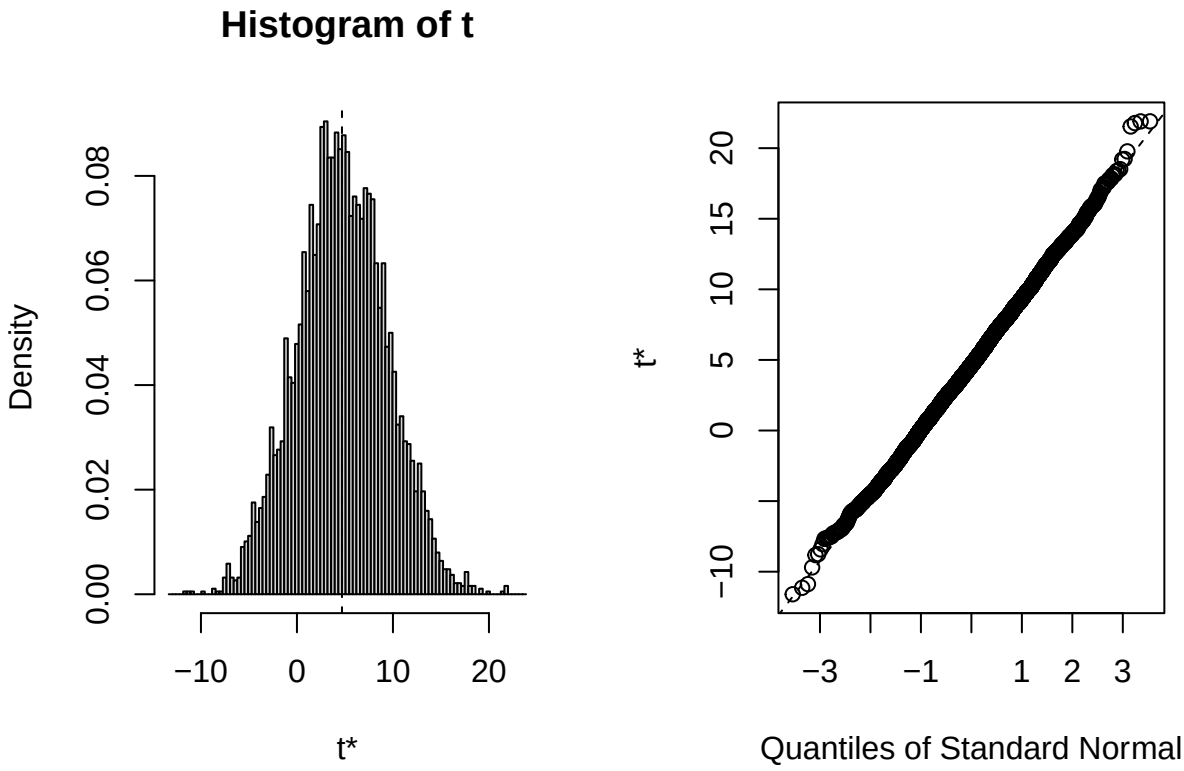
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_CU, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-9.0914,  8.8950 )
## Calculations and Intervals on Original Scale

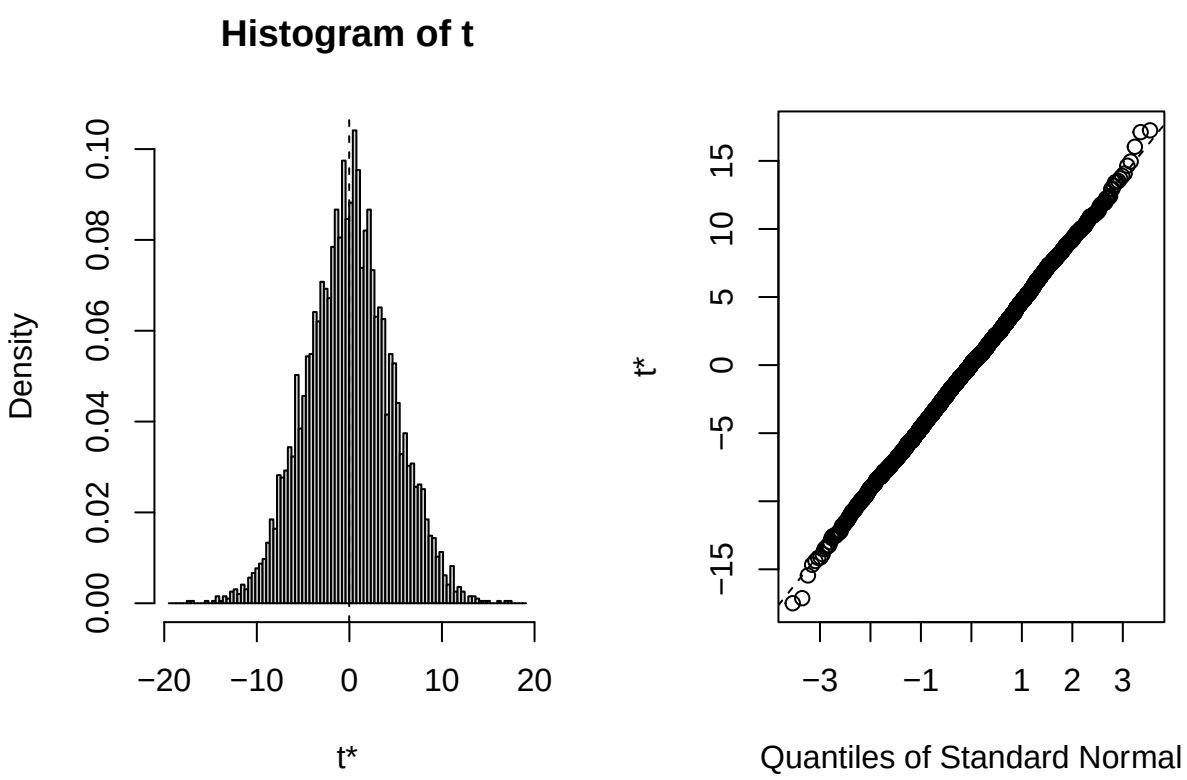
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_CU, conf = 0.99, type = "bca",
##     index = 3)
##
## Intervals :
## Level       BCa
## 99%   (-6.667, 17.500 )
## Calculations and Intervals on Original Scale
```

4.3.3) Model diagnostics






Histogram of t
Density
0.00
0.02
0.04
0.06
0.08
−10
0
10
20
t*
t*
−10
0
5
10
15
20
−3
−1
1
2
3
Quantiles of Standard Normal




```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_CU, conf = 0.99, type = "bca",
##     index = 1)
##
## Intervals :
## Level       BCa
## 99%   (-5.37, 29.32 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_CU, conf = 0.99, type = "bca",
##     index = 2)
##
## Intervals :
## Level       BCa
## 99%   ( 0.3885,  0.9779 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
```

```
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_CU, conf = 0.99, type = "bca",
##     index = 3)
##
## Intervals :
## Level       BCa
## 99%   (-6.667, 17.500 )
## Calculations and Intervals on Original Scale

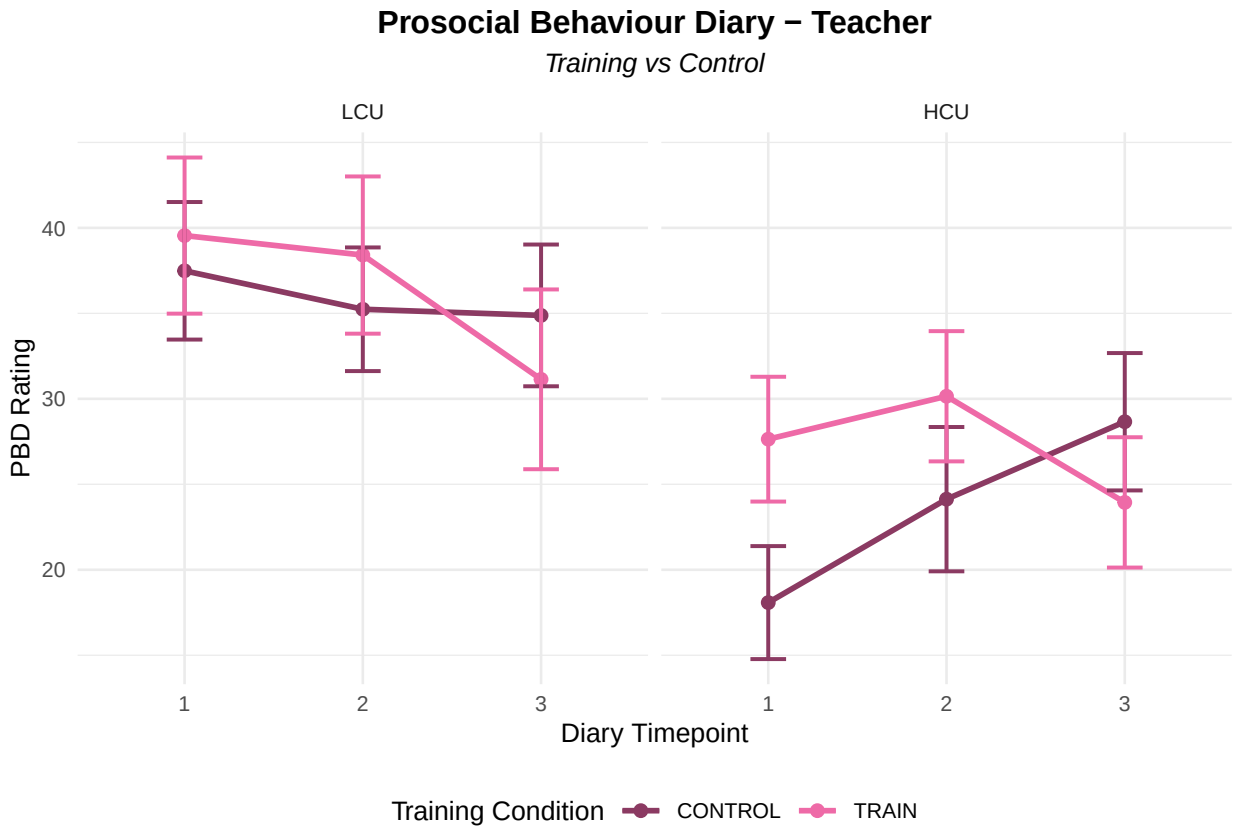
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_CU, conf = 0.99, type = "bca",
##     index = 4)
##
## Intervals :
## Level       BCa
## 99%   (-12.3389,  11.3202 )
## Calculations and Intervals on Original Scale
```

4) Teacher Prosocial at 2 weeks

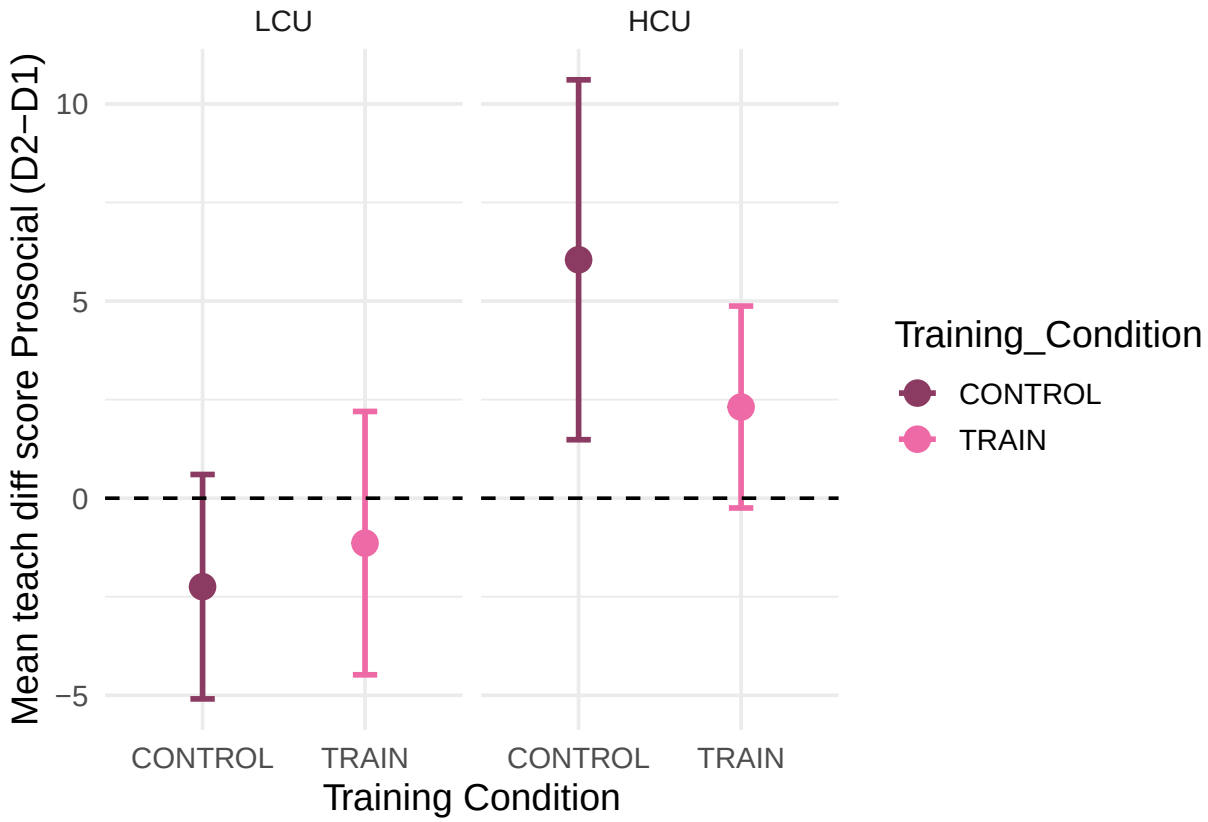
4.1) Table of outcomes

Training_Condition	diaryTP	CP_Group	Group_N	Mean_ABD	se_ABD	Mean_PBD	se_PBD
CONTROL	1	LCU	32	14.68	3.38	37.49	4.023299
CONTROL	1	HCU	21	17.39	3.30	18.08	3.305231
CONTROL	2	LCU	32	17.70	3.16	35.24	3.616988
CONTROL	2	HCU	21	18.62	3.54	24.13	4.220790
CONTROL	3	LCU	30	15.06	2.26	34.88	4.146050
CONTROL	3	HCU	21	17.02	3.35	28.66	4.017480
TRAIN	1	LCU	25	21.24	4.12	39.55	4.566927
TRAIN	1	HCU	25	21.40	3.94	27.64	3.650338
TRAIN	2	LCU	25	22.10	2.90	38.41	4.598401
TRAIN	2	HCU	25	16.68	3.26	30.15	3.809410
TRAIN	3	LCU	21	25.85	4.10	31.14	5.258720
TRAIN	3	HCU	24	23.92	3.70	23.94	3.811794

4.2) Plot of outcomes



Difference plot:



4.3) Linear Regression

```
model_t_psb_CU_lm <- lm(PBD_mean_rating ~ Mean_PBD_D1 + Training_Condition + CP_Group +  Training_Condi
                      data =  teach_pbd_d3_only)

model_t_psb_CU_lm_nointeraction <- lm(PBD_mean_rating ~ Mean_PBD_D1 + Training_Condition + CP_Group,
                      data =  teach_pbd_d3_only)
```

4.3.1) Does interaction term improve model? No.

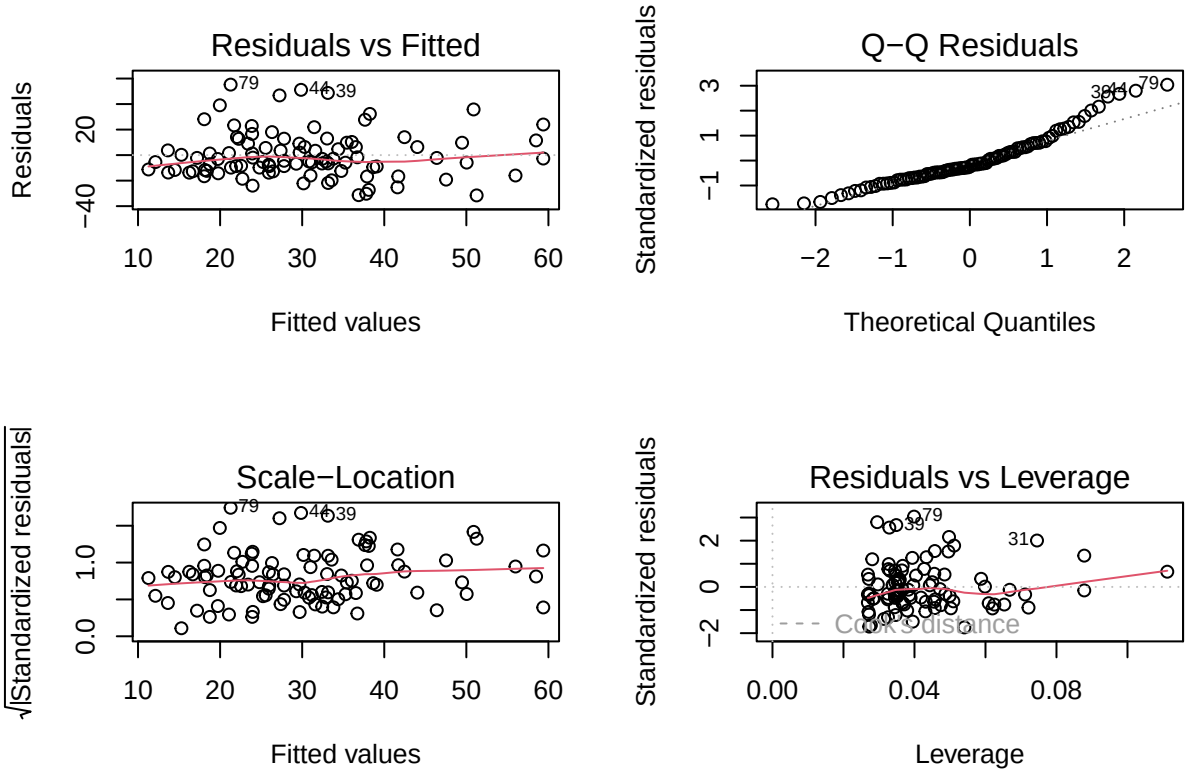
```
## Analysis of Variance Table
##
## Model 1: PBD_mean_rating ~ Mean_PBD_D1 + Training_Condition + CP_Group +
##     Training_Condition * CP_Group
## Model 2: PBD_mean_rating ~ Mean_PBD_D1 + Training_Condition + CP_Group
##   Res.Df   RSS Df Sum of Sq      F Pr(>F)
## 1     90 31134
## 2     91 31288 -1   -154.43 0.4464 0.5058
```

4.3.2) Model summary

```
##
```

```
## Call:
## lm(formula = PBD_mean_rating ~ Mean_PBD_D1 + Training_Condition +
##     CP_Group, data = teach_pbd_d3_only)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -31.590 -12.015  -4.460   9.346  55.217
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)              16.24882    4.63590   3.505 0.000711 ***
## Mean_PBD_D1               0.50735    0.09503   5.339 6.83e-07 ***
## Training_ConditionTRAIN -6.79665    3.87799  -1.753 0.083035 .
## CP_GroupHCU               1.82062    4.13275   0.441 0.660595
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 18.54 on 91 degrees of freedom
## Multiple R-squared:  0.2642, Adjusted R-squared:  0.2399
## F-statistic: 10.89 on 3 and 91 DF,  p-value: 3.52e-06
```

4.3.3) Assumpation test



4.4) Bootstrapped Linear Regression

4.4.1) Results and specification

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = teach_pbd_d3_only, statistic = boot_coef_t_psb_cu,
##     R = 5000)
##
##
## Bootstrap Statistics :
##       original       bias    std. error
## t1* 16.2488242  0.166122890  5.31951753
## t2*  0.5073544 -0.003633485  0.09703688
## t3* -6.7966501 -0.041004425  3.66450925
## t4*  1.8206196 -0.004073813  3.96601801
```

4.4.2) Confidence Intervals

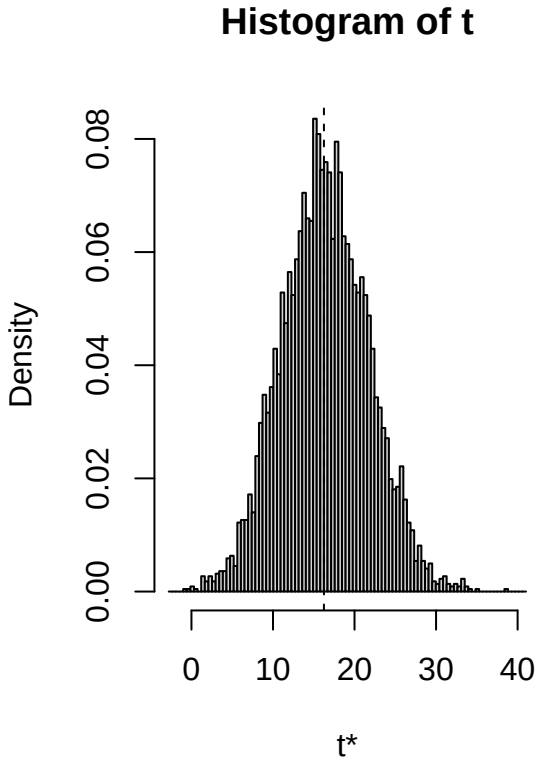
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb_cu, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   ( 6.68, 27.25 )
## Calculations and Intervals on Original Scale

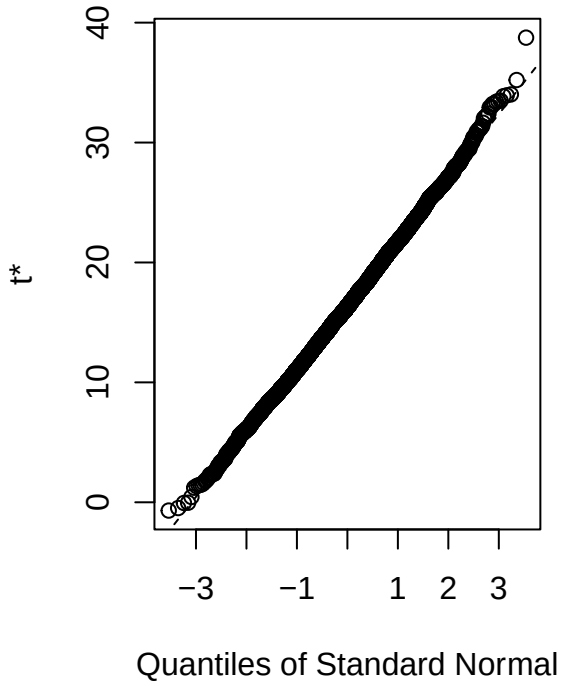
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb_cu, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.3102,  0.6861 )
## Calculations and Intervals on Original Scale

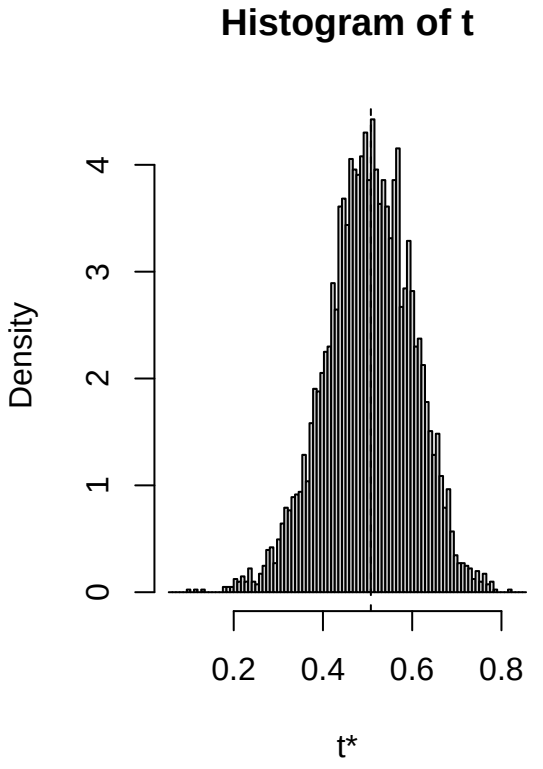
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb_cu, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-14.027,   0.291 )
## Calculations and Intervals on Original Scale
```

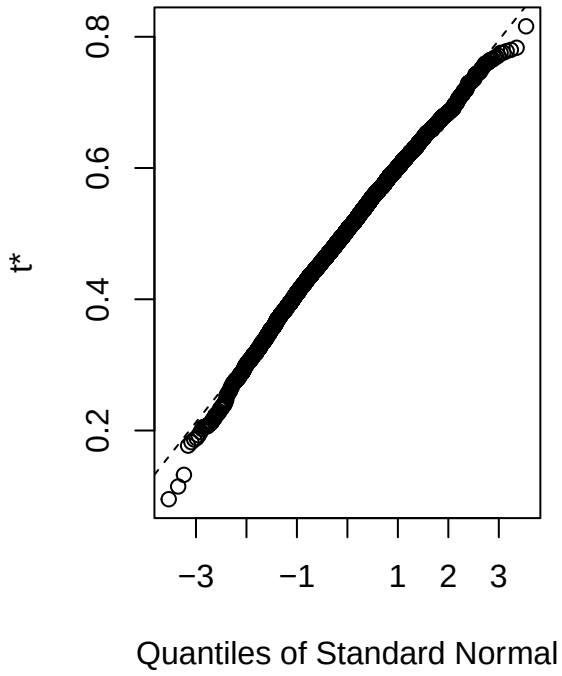
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb_cu, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-5.979,  9.721 )
## Calculations and Intervals on Original Scale
```

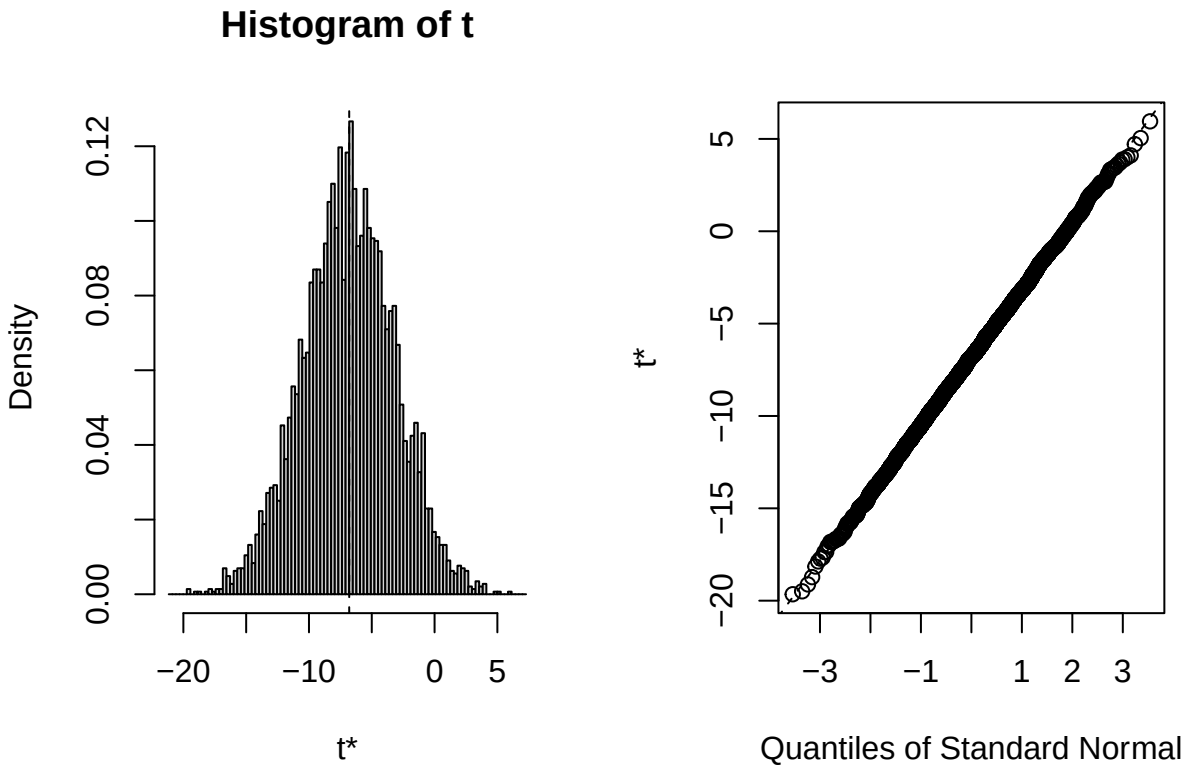
4.4.2) Model diagnostics

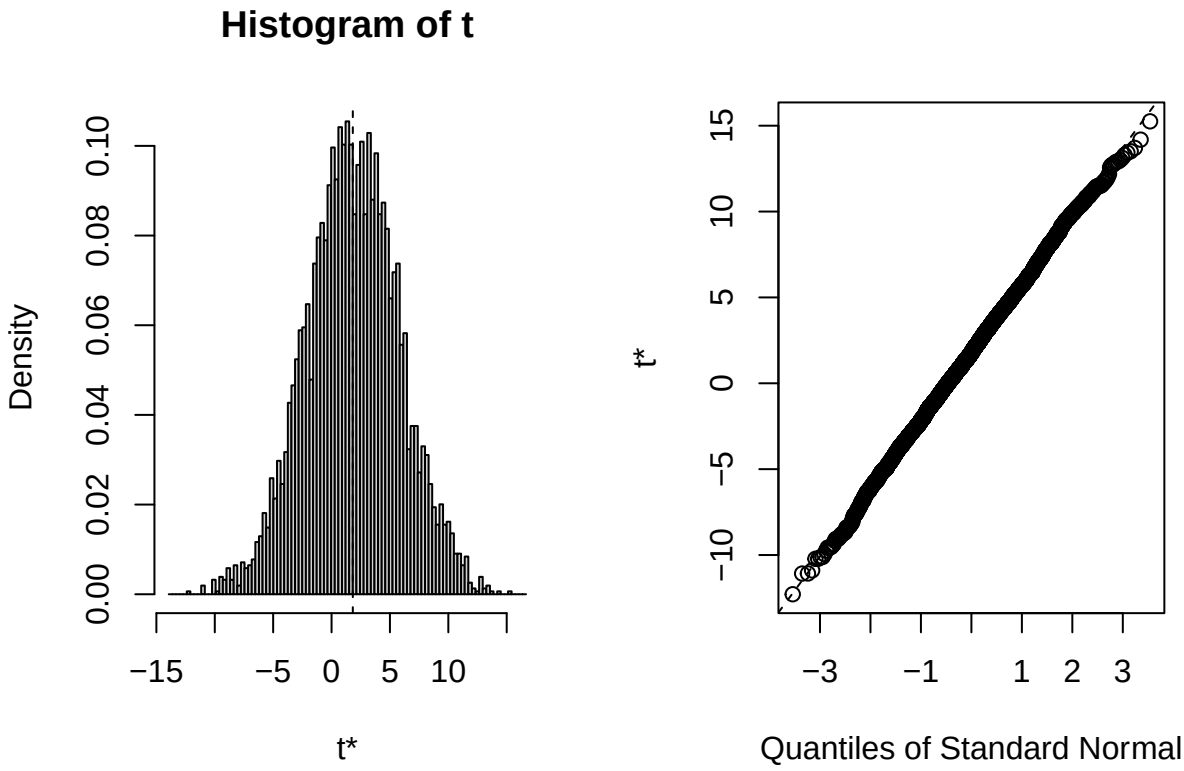









Histogram of t
Density
0.00
0.04
0.08
0.12
−20
−10
0
5
t*
t*
−20
−15
−10
−5
0
5
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.02
0.04
0.06
0.08
0.10
−15
−5
0
5
10
t*
t*
−10
−5
0
5
10
15
−3
−1
1
2
3
Quantiles of Standard Normal